# Development of an On-Site Formwork Material Inventory, Matching and Cutting Optimization System for Two-Storey Residential Buildings

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# Chapter 1

# INTRODUCTION

## 1.1 Background of the Study

The construction industry generates a significant amount of material waste through cutting, installation, and material handling that is deposited on the construction site (Bandara, Waidyasekara & Ranasinghe, 2026). According to Liu, Yi & Wang, (2020), construction waste reduction is influenced by practices within construction projects, indicating that waste reduction should be addressed through better material management and construction practices. Similarly, Ajayi et al., (2017) identified material reuse, waste segregation, and logistics management as important practices for minimizing construction waste on site. These findings suggest that construction waste reduction should not focus only on disposal but also on improving how materials are identified, stored, and reused during construction.

Offcuts are a common material loss during construction because materials are frequently cut to meet specific dimensions. This is particularly applicable to formwork involving wood waste and any carpentry involved. Plywood used for formwork may be cut to suit slab, column, and beam dimensions. Including lumber and other wood-based materials used for temporary construction works may generate offcuts and smaller pieces during cutting and installation. Cutting optimization can help minimize these losses by planning cutting patterns according to the required dimensions and available material sizes. Mei et al. (2022) developed a BIM-based formwork planning framework that used a cutting-stock optimization model to determine cutting

patterns and reduce the quantity of formwork materials required. However, cutting optimization does not eliminate all offcuts, making the identification and potential reuse of remaining pieces relevant to construction material management. These remaining pieces may be treated as waste when they are no longer suitable for their original purpose. Kern, Amor, Angulo & Montelongo, (2018) found that temporary wood use in construction can represent a significant portion of construction waste, with design and production characteristics influencing the amount of wood waste generated.

The issue is particularly evident in plywood formwork Bandara et al., (2026) found that offcuts generated during cutting operations were the major source of plywood waste in the projects they investigated. The study also found that plywood could generally be reused for approximately three to five cycles depending on material quality, supervision, and handling. Importantly, some plywood with minor damage could still be trimmed and reused for smaller construction applications. These findings show that material remaining after construction activities should not automatically be considered unusable waste because its dimensions and condition may still allow further applications.

The importance of systematic material recording is demonstrated by Yu et al., (2023), who proposed a sustainable building design and construction framework for the reuse of irregular off-cut wood. Their Material Reclamation System records information about reclaimed materials and organizes the information into a database. The database can include characteristics such as material type, dimensions, weight, age, and material grade. The purpose of this approach is to make recovered materials easier to identify and match with potential applications. Yu et al., (2023) also distinguished that fewer than 20% of the reviewed reclaimed wood construction projects had established databases

for recycled materials, which indicates a gap in systematic material recording and reuse.

However, the mentioned studies generally focus on construction waste reduction, material reuse, or specific material types. Without a centralized inventory system, these materials may be managed through workers' memory, visual inspection, or informal storage. This makes it difficult to track available materials, number of uses, dimensions, and potential for reuse. The proposed system will focus on recording and organizing information about usable formwork materials during construction. The inventory may include type of material, dimensions, quantity, condition, and possible reuse application.

## 1.2 Objective of the Study

The general objective of this study is to develop an on-site formwork cutting optimization and recoverable materials inventory utilization system for two-storey residential buildings to improve recording, monitoring, potential utilization, potential procurement savings and cutting waste reduction of formwork materials.

The specific objectives of this study are:

1. To reduce offcut waste by optimizing the cutting of formworks
2. To classify materials according to dimension, quantity, number of uses and physical condition
3. To develop a prototype of an on-site material optimization inventory and matching system that records reusable formwork materials and determines optimized cuts for formworks provided
4. To determine accuracy in recording and matching of materials inventory to required materials for formworks

5. To determine potential procurement savings from the proposed system

## 1.3 Significance of the Study

### To Construction Industry

The study may contribute to a systematic way of recording and identifying reusable formwork materials in residential construction. By organizing information on available reusable materials and optimizing cutting patterns, it may support waste reduction practices and encourage the consideration of maximizing the materials before new materials are purchased.

### To Contractors and Developers

The developed application may help contractors and developers manage formwork materials more efficiently by providing organized information about their availability and optimization in cutting. This may help reduce unnecessary material purchases and improve the utilization of materials already available on the construction site.

### To Site Engineers and Project Engineers

The application may serve as a practical reference for site and project engineers when looking for available formwork materials for specific construction needs and also provide optimized cutting patterns. By searching for the recorded inventory, engineers can identify potentially suitable materials without having to manually inspect or recall all stored materials and inputting dimensions of phenolic plywood and identifying the optimized cutting dimensions for this.

### To Construction Workers

The developed application may provide construction workers with a convenient reference when looking for materials needed for their assigned tasks

and also a reference for optimized cutting of formworks. By allowing them to search for available formwork materials based on their material requirements, the system may reduce unnecessary searching and handling of stored materials while helping them identify formwork materials that may still be useful. It may also encourage workers to properly store and report usable materials for future construction activities.

**To Future Researchers**

The study may serve as a reference for future researchers interested in construction material management, construction waste reduction, inventory systems, and digital application. It may also provide a basis for further studies involving other construction materials, additional utilization criteria, cutting optimization or more advanced recommendation features.

## 1.4 Scope and limitation

The study will focus on developing an onsite formwork materials cutting optimization and inventory utilization system for two-storey residential buildings. The developed system will be tested in selected 3 two-storey residential buildings. The materials that will be focused for this study are lumber and phenolic board. The system will be a Progressive Web App (PWA) which can be used offline and online. The system can run across all devices. The system will determine optimized cutting for formwork materials, suggest possible uses of recoverable materials and record their material type, dimensions, quantity, condition and number of uses. The developed system will be designed primarily as a tool for optimizing, inventory and monitoring of formwork materials onsite. The system will allow users to enter dimensions of needed material and the system will automatically identify and suggest cutting plan for new formwork materials, available material in the inventory that matches the required material

type and dimensions will be suggested for use. The dimensions of the recovered materials will be measured manually and the conditions will be based on the inspection of materials. The performance of the system will be evaluated based on the accuracy of system matching of reusable materials, waste reduction and potential procurement savings from optimization based on the dimensions required. The study will be limited to selected two-storey residential building construction projects. The study will not focus on all types of construction wastes. Materials that are severely damaged, irregularly shaped or unsuitable for reuse will not be included in the inventory and utilization system. The system will only suggest optimized cutting for formwork materials, and possible recoverable materials that can be used for the given dimensions, the use of the potential materials shall be subjected to review and approval of an onsite in charge civil engineer or personnel.

## 1.5 Conceptual Framework

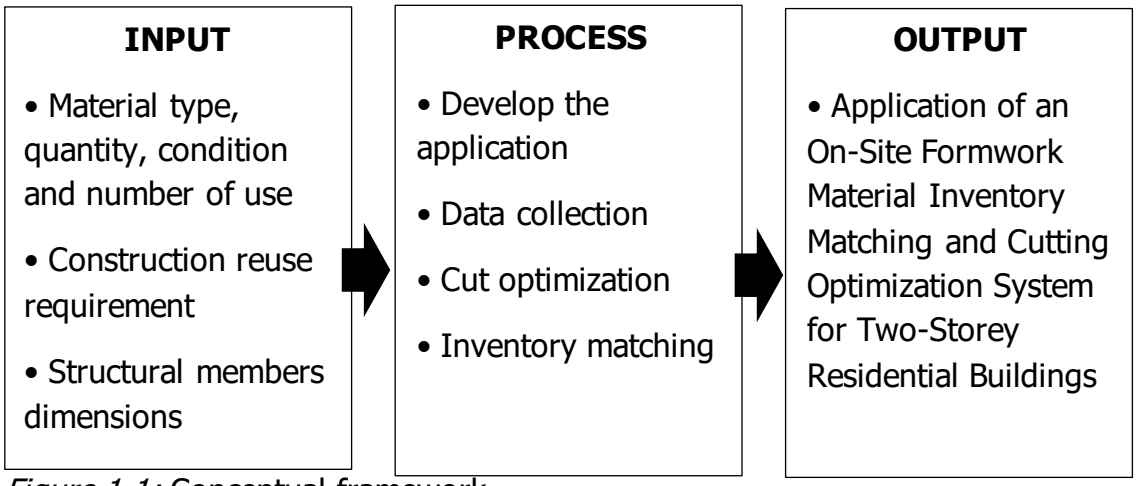



*Figure 1.1:* Conceptual framework

## 1.6 Definition of Terms

Centralized inventory system - A cloud-based digital system that gathers and organizes material information, stock levels, and transaction records in

one centralized online platform. It allows authorized users to access and update inventory data in real time, regardless of their location, providing consistent records and efficient monitoring of materials across the organization.

Condition - This refers to the physical state of recoverable material, such as its level of damage, usability, or suitability for potential reuse.

Cutting optimization – The process of determining an arrangement or sequence of cuts that satisfies required piece dimensions while reducing stock material consumption, and residual waste

Inventory - This refers to the organized record of available formwork offcuts and recoverable materials, including their type, dimensions, quantity, condition, and storage location.

Offcut - This refers to the portion of a construction material left after the material has been cut to the required size or shape for a particular construction activity.

Reclaimed wood construction projects - Reclaimed wood is a sustainable material with a renewable and unique customized nature, and the reclaimed wood esthetics and urban wood esthetics that represent nature make recycled wood products stand out from the competition in recycled products

Two-Storey Residential Building - This refers to a residential structure consisting of two above-ground floors and serves as the building type covered by the study.

Progressive Web App - is a type of web application that works like a regular website but can provide features similar to a mobile or desktop app.

# Chapter 2

# REVIEW OF RELATED LITERATURE

## 2.1 Construction Material Waste in Construction Industry

Construction material waste refers to materials that are unused, discarded, damaged and lost during construction activities. Construction material waste is potentially generated during preparation, cutting, installation, and dismantling. In line with the exponential growth of the population, the requirement for urban expansion develops and further results in the generation of massive amounts of construction waste (Luangcharoenrat, Intrachooto, Peansupap & Sutthinarakorn, 2019), causing an unbalanced ecosystem and generation of large volumes of construction material waste (Afolabi, Olayeni, Ojelabi & Omuh, 2018). The rise of resource consumption in the construction industry leads to a concerning amount of natural resources degradation, these concerns require a shift to sustainable construction through a circular economy (Hossain, Ng, Afari & Amor, 2020). According to Benachio, Freitas & Tavares, (2020) Construction industry has a significant role in the development of circular economy because construction activities are responsible for large consumption of natural resources and massive generation of solid wastes. The study emphasizes ways such as reuse of material and recycling waste to improve and maximize the value of the materials. Ginga, Ongpeng & Daly, (2020) address that construction and demolition waste may contain materials that can potentially be recovered, reused and recycled instead of immediately disposing after use.

*Table 2.1:* Percentage of waste from construction compared with total construction waste.

| Country | Percentage of Waste | References |
|---|---|---|
| England | 32% | Sharman [5] |
| Hong Kong | 28% | EPDHK [6] |
| Netherland | 28% | |
| Australia | 20–30% | |
| United States | 20–29% | Bossink and Brouwers [7] |
| Germany | 19% | |
| Finland | 13–15% | |
| Chili | 34% | Mager [8] |
| Brazil | 50% | Contreras et al. [9] |
| Denmark | 27% | Katz and Baum [10] |
| Israel | 60% | Allwood et al. [11] |
| Japan | 20% | Yonetani [12] |
| Canada | 27% | Yeheyis et al. [13] |

*Source:* (Luangcharoenrat et al., 2019)

### 2.1.1 Common causes of material waste on construction sites

Construction material waste can occur because of the different activities and conditions of the construction site (Luangcharoenrat et al., 2019). Design changes, design error, inaccurate estimation of materials, cutting losses and shipping errors are some of the contributors to the generation of material wastes in the construction industry Bachayo, Memon, Hussain, Rahman & Ahmed, (2022). Similarly, Kaliannan, Nagapan, Abdullah, Sohu & Jhatial, (2018) supported by Mahamid, (2024) major causes of construction wastes generation are design changes, improper storage, poor material handling, weather conditions, and incorrect ordering of materials. Their findings prove that construction material wastes are influenced not only by the material itself but also by the handling, planning and management of materials in the actual

construction site. These studies imply that reducing material waste requires better management throughout the whole construction process. Materials should be properly quantified, stored, handled, and monitored.

### 2.1.2 Construction and demolition waste

Construction and demolition wastes are wastes that are generated during construction activities, it contributes to large volumes of wastes received by landfills which then lead to major environmental issues (Polat, Damci, Turkoglu & Gurgun, 2017).

*Table 2.2:* Rank of factors that may create C&D waste.

| C&D waste factor group | C&D waste factors | Ave. Rat.* | RII† | Ranking by Cate. | Overall Ranking | Importance Level |
|---|---|---|---|---|---|---|
| Design and contract documents | Errors in contract documents | 2.88 | 0.58 | 5 | 21 | M |
| | Design and construction detail errors | 4.00 | 0.80 | 2 | 2 | H |
| | Design and detailing complexity | 2.97 | 0.59 | 4 | 19 | M |
| | Frequent design changes and change orders | 4.14 | 0.83 | 1 | 1 | H |
| | Selection of low quality materials | 3.55 | 0.71 | 3 | 10 | H-M |
| Procurement | Purchasing materials not complying with specifications | 3.17 | 0.63 | 3 | 15 | H-M |

*Source:* (Polat et al., 2017)

The study of Ginga et al., (2020) highlights that construction and demolition waste should not be viewed only as a disposal problem but also as potential source of recoverable material. The study emphasizes construction and demolition waste materials recovery, reuse, recycling and circular economy as important ways for reducing the environmental impact of construction activities. Similarly, Kabirifar, Mojtahedi, Wang & Tam, (2021) explained that construction and demolition waste is a serious concern since ineffective management of construction waste may pose environmental and economic consequences throughout the whole construction process. The study emphasizes the importance of preventing waste and reusing materials should be considered.

Ruiz, Ramón & Domingo, (2020) reviewed circular economy strategies to mitigate construction and demolition waste leading to minimization of wastes and reduction environmental implications. Their study highlights the importance of solutions before materials eventually become wastes, supporting the concept of waste prevention and reuse of materials should be implemented in the construction process. Similarly, a local study of Estrada, Nakatani, Hayashi & Fujita, (2023) emphasizes that construction and demolition wastes must be maximized rather than disposing them to landfills. For residential construction projects, these concepts are applicable to lumber and formwork materials. Materials removed after use in formwork applications may not immediately go to waste. Depending on its dimensions, physical condition and number of reuses it may remain suitable for another formwork application. Therefore, ineffective management of possible reusable materials could result in loss of potential reusable resources.

### 2.1.3 Material wastes generated in construction sites

Construction sites generate various types of material wastes from excess materials, defective materials, damaged materials, offcuts, and leftovers after completion of construction activities. One of the relevant categories is offcuts, since materials that are used for construction activities undergo cutting to achieve specific dimensions. The study of Roy & Beaulah, (2025) identified that the core source of material waste is material offcuts, followed by packaging and excess materials. The study also highlighted that due to fast growth of the construction industry, it leads to additional generation of waste. There is an immediate need for advancement in waste reduction to achieve sustainable construction. Wang & Yi, (2022) studied specifically focused on offcuts generated in construction. Mahamid, (2024) found that design errors, rework, lack of labor skills, design

changes and non-compliance with the specification contributes to material wastes generated for construction projects.

### 2.1.4 Construction material waste in small scale and residential projects

Small scale construction projects pose more challenges in construction management compared to large scale construction projects because of limited storage spaces and reliance on manual material management. The study of Kittiwararat & Lertwattanaruk, (2019) focused on architectural wastes generated on small scale residential construction projects. They then found that appropriate material management and construction project planning could reduce generated wastes. Residential building projects also involve repeated concrete activities which may require the repeated use of formwork for structural members such as columns, beams, slabs, stairs, and foundations. This repeated activity creates opportunities for material reuse but only if materials can be identified properly. Maués, Nascimento, Lu & Xue, (2020) studied wastes generated in construction of residential buildings. They concluded that floor area and numbers of floors are significant in waste volume generation for residential buildings. In the study of Wang, Chen, Hu & Ren, (2019) waste generation from residential building construction were studied. It was found out that the amount of waste varies among construction projects which further reinforced the need for monitoring of material during the construction phase of the project. Similarly, the study of Estrada et al., (2023) conducted in Manila, Philippines investigated the construction and demolition waste generation in residential projects. The construction and demolition waste generation were determined and evaluated, It is then identified the need for sorting of waste which could improve the waste management and increase opportunities for recovered materials that can be potentially utilized. Design changes, rework, procurement and labor skills are

factors that affect material wastes for residential building projects which requires coordination of every stakeholder in construction to reduce waste generation (Mahamid, 2024).

### 2.1.5 Common construction wastes

Eze, Seghosime, Eyong & Loya, (2017), in their study on material wastage from the perspectives of construction operatives, tradesmen, and artisans in Nigeria, identified wood and timber formwork, mortar from rendering and plastering, and blockwork or brickwork as the most wasteful materials reported by the respondents. The study also identified poor material storage, design changes, theft, and vandalism as factors contributing to material waste. These findings indicate that formwork is a relevant material category to consider when examining construction waste and developing strategies for material management. Similarly, (Rahaman, Samad & Taib, 2017) reviewed construction methods for minimizing waste in Malaysia and identified wood as a major waste material associated with the conventional formwork system. Their discussion highlighted the use of plywood and engineered wood in formwork and the potential of alternative construction methods to reduce waste generation. Together, these studies establish the relevance of wood-based formwork in construction-waste reduction. (Bakchan & Faust, 2019) also examined construction waste generation in an institutional building project using 535 waste-hauling tickets. Their results showed that concrete and masonry were the principal waste contributors during the foundation and structural concrete stages, whereas wood became the primary contributor during masonry and finishing. Across the project, concrete and masonry accounted for approximately 48.5% of the total waste, while wood accounted for approximately 41%. These findings demonstrate that material waste composition changes as construction progresses. In particular, wood waste can become a major contributor during

specific stages, even when it does not rank first across the entire project. This supports the importance of considering both the material category and the construction stage when prioritizing waste-management measures. Furthermore, (Luangcharoenrat et al., 2019) examined factors influencing construction waste generation in Thailand. Although their study focused on waste-generation factors rather than ranking material quantities, it identified design changes, inattentive working attitudes, ineffective planning and scheduling, and improper material storage as significant contributors to waste generation.

## 2.2 Direct Reuse Potential

The identification of common construction material waste provides the initial basis for determining materials that may be considered for recovery and reuse. However, frequency of a material result in meta-pareto analysis does not automatically indicate that it can be directly reused. Some construction materials can be reused directly retaining the same form while other materials require processing or transformation before they can be utilized for another construction application. Since the proposed study aims to develop an on-site inventory and matching system for materials that can be reused, the difference between direct reuse and recycling is important.

### 2.2.1 Direct reuse versus recycling

Direct reuse and recycling are different approaches to material recovery in the circular economy. Direct reuse involves using an existing product or component again without transforming it into a different product through a reprocessing procedure. Recycling involves reprocessing waste materials to produce new materials that may serve the same or different purpose. The primary distinction between these approaches is whether the existing product or component is used again directly or whether the waste undergoes reprocessing

before it can be utilized again. In the study of (Kirchherr, Reike & Hakkert, 2017) they emphasized that reuse and recycling are different activities towards a circular economy. Their study emphasized the importance of these strategies for circular economy, while both are strategies toward circular economy they involve different processes on recovering materials.

### 2.2.2 Direct reuse potential of common construction material

Construction materials such as concrete components, bricks, structural steel, timber, doors, and windows may retain value after their original use and can potentially be recovered for subsequent construction activities. (Ginga et al., 2020) reviewed material recovery from construction and demolition waste and identified differences in the reuse potential of various construction components. Their review indicated that structural timber, structural steel, clay bricks bonded with lime-based mortar, and concrete paving slabs may have relatively high reuse potential, while in-situ concrete generally has lower potential for direct reuse. These differences demonstrate that the possibility of direct reuse depends on the material type, its original installation, and whether it can be recovered without substantial damage. Similarly, (Küpfer, Bastien-Masse & Fivet, 2023) reviewed 77 examples of concrete component reuse in new construction projects. Their study demonstrated that concrete pieces can sometimes be extracted from existing structures and reused in new assemblies without first crushing them into recycled aggregates. However, this approach remains less common than conventional concrete recycling and requires appropriate recovery and assessment procedures. This distinction is important because the reuse of intact concrete components preserves more of their original form, whereas producing recycled aggregates requires the concrete to be broken down and processed.

#### *2.2.2.1 Direct Reuse Potential of Concrete*

Concrete waste has potential for reuse but cannot be directly utilized as it needs to undergo processing before it can be utilized in new construction products. According to (Ginga et al., 2020), material recovery involves processing construction and demolition waste to facilitate its conversion into useful materials. For concrete waste, common procedures include crushing, screening, sorting, and removing contaminants to obtain recycled aggregates with suitable particle sizes and characteristics. (Mandal & Shiuly, 2025) discussed how the properties of recycled concrete aggregates depend on their composition and processing. Therefore, the recovery process must be appropriate for the intended application to ensure that the resulting material meets relevant construction requirements. Concrete waste cannot generally be used directly as a substitute for new concrete because its physical condition, composition, and mechanical properties may be unsuitable for the intended application. Demolished concrete may contain attached mortar, cracks, irregular fragments, and contaminants that affect its quality. (Mandal & Shiuly, 2025) explained that recycled concrete aggregates may exhibit higher water absorption, greater porosity, and different mechanical properties than natural aggregates. They reported that 100% recycled aggregate content was associated with roughly a 30% decrease in compressive strength in the reviewed data. These characteristics can influence the strength, workability, and durability of newly produced concrete, making material processing and quality assessment necessary before utilization.

#### *2.2.2.2 Direct Reuse Potential of Steel*

Steel reinforcement is another material that generates offcuts during construction because reinforcing bars are supplied in standard lengths and must

be cut to the lengths required by the structural drawings. (Lee, Son, Kim & Kim, 2020) noted that rebar cutting waste is typically estimated at 3-5% during planning, while more than 5% occurs during actual construction. (Widjaja & Kim, 2023) similarly reported site cutting waste ranging from 5–8%. In response, rebar cutting waste has long been treated as a cutting stock problem, and (Kwon, Kim & Kim, 2021) reviewed 52 studies on the topic, concluding that although many optimization algorithms have been proposed, a small amount of cutting waste is still generated in practice. A cutting waste minimization process is described by (Lee et al., 2020). The process reads the rebar cutting list, applies options such as the minimum and maximum order lengths, target loss rate, and minimum quantity, and then performs minimization by special length first. Minimization by stock length is performed on the remaining rebar. Applied to a commercial building project, the algorithm produced a cutting waste of 0.58% for 25 mm bars combined by special length compared with 1.58% by stock length. The overall loss rate across all bar diameters was 0.96%, and the quantity of rebar was reduced by 6.04% compared with the actual quantity used. These findings indicate that steel waste is addressed mainly at the planning stage, before cutting takes place, rather than through the reuse of leftover pieces. Because optimization leaves very little residual material, the pieces that remain are generally short. (Rácz, 2025) defined trim loss as the leftover that is not long enough to be reused in the future, and in the proposed model a leftover shorter than a set threshold is counted as waste. Only longer leftovers therefore retain value. (Widjaja & Kim, 2023) likewise noted that rebar is more challenging to reuse than materials such as concrete and bricks. (Lee et al., 2020) observed that leftover small-diameter bars may still serve secondary purposes, such as diagonal bars for crack reinforcement or reinforcement around openings, but this is a minor use and not direct reuse for the original application.

*2.2.2.3 Direct Reuse Potential of Formwork*

Formwork materials provide another opportunity for direct reuse because they are temporary materials used to shape concrete during construction. Once the concrete has gained sufficient strength, formwork is removed and can be used again if it remains suitable for the next concreting activity (Balasbaneh, Sher & Ibrahim, 2024). Formwork is relevant to the system because their dimensions and physical condition can be inspected and recorded (Yu et al., 2023). In addition, (Mei et al., 2021) highlighted the importance of considering formwork dimensions and reuse requirements when planning formwork utilization. The study of (Bandara et al., 2026) examined plywood usage and wastage, they reported that plywood formwork was reused for approximately three to five cycles depending on material quality, supervision and handling practices. The study also identified offcuts generated during cutting operations as major source of plywood waste. Their study of reuse potential, formwork materials are suitable for an inventory and matching approach because they can be characterized using measurable and observable information such as material type, length, width, thickness, quantity, physical condition and number of previous uses. These data can be recorded in the proposed system and compared with the dimensions and quantity required for subsequent formwork activities. This approach is further supported by (Pronk, Brancart & Sanders, 2022), who investigated the reuse of timber formwork in building construction, demonstrating the potential to give recovered timber a subsequent application rather than immediately treating it as waste. In addition, the study of (Apdeni, Hellyward & Syihab, 2024) emphasized that in a building construction project, formwork is seen as one of the most

significant material based on meta-pareto items in sustainability for structural work together with concrete and steel.

## 2.3 Construction Offcuts and Excess Material

Construction activities involve the cutting, sizing, installation, and adjustment of materials according to project requirements. These processes can generate leftover pieces or quantities of materials, referred to as construction offcuts and excess materials. These may include wood, lumber, plywood, metal, plastic, and other materials used in construction (Benachio et al., 2020). The resulting offcuts can vary in material type, dimensions, quantity, shape, and physical condition. Yu et al. (2023) noted that offcut wood can have different physical characteristics depending on how materials are processed.

### 2.3.1 Offcuts from temporary construction activities

Temporary construction activities involve the preparation, cutting, adjustment, and installation of materials according to project requirements. These processes can generate leftover materials and offcuts in varying sizes and quantities. (Bandara et al., 2026), in their study of plywood usage and wastage in slab and beam construction, identified cutting as one of the major sources of plywood offcuts and noted that material wastage can be influenced by factors such as structural geometry, material dimensions, handling practices, and construction site conditions. These findings indicate that the characteristics and quantity of offcuts generated during construction can vary depending on the requirements and conditions of the construction activity.

#### *2.3.1.1 Generation of Wood Offcuts*

Wood offcuts are generated when wood-based construction materials are cut, trimmed, or adjusted to meet the dimensions required for specific construction activities. These processes produce leftover pieces that can vary in length, width, thickness, shape, and quantity depending on the material and the required dimensions. Jahan, Zhang, Bhuiyan, & Navaratnam (2022) identified offcuts, sawdust, slabs, and other wood residues as forms of wood waste that can result from construction and related processing activities. The generation of wood offcuts is influenced by the preparation and dimensional adjustment of wood-based materials. Wu, Yu & Poon (2019) examined construction waste composition and identified wood and timber as among the waste materials generated on construction sites. Their findings demonstrate that wood waste can be generated as part of construction activities and can occur in different quantities depending on the materials and processes involved. Yu et al. (2023) discussed off-cut wood generated when wood materials are cut according to required dimensions. The study described offcuts as pieces having varying dimensions and geometries, which result from the cutting and preparation of wood materials which lead to generation of wastes. These findings indicate that the characteristics of wood offcuts are related to the generation of wastes in construction activities. For the present study, the generation of wood offcuts will be considered in relation to two-storey residential building construction.

#### *2.3.1.2 Offcuts from Lumber and Other Wood-Based Materials*

Lumber, plywood, and other wood-based materials differ in their dimensions, thickness, composition, and physical properties, which can influence the characteristics of the leftover pieces produced during construction activities. Yu et al. (2023) documented off-cut wood according to characteristics such as

wood type, length, width, weight, and material proportions. These variations can result from differences in material dimensions and cutting requirements. Timber offcuts are also identified as a component of construction of timber waste. The occurrence of these offcuts is associated with the processing and dimensional adjustment of wood-based materials during construction activities.

### 2.3.2 Offcuts from formwork activities

Formwork is a temporary construction system used to provide the required shape and support for fresh concrete until it gains sufficient strength. Its preparation and installation involve activities such as measuring, cutting, fitting, assembling, erecting, and adjusting materials according to the dimensions and geometry of structural elements. Li , Lin, Bao & Xie (2022) explained that formwork systems are prepared according to the requirements of concrete structures and may involve different materials and fabrication methods. These activities require formwork components to be prepared according to the dimensions of slabs, beams, columns, walls, and other concrete elements, which can result in leftover pieces when materials are cut or adjusted. Wood-based materials are commonly used in formwork activities. Li et al. (2022) identified wood as one of the traditional materials used in formwork systems and discussed the fabrication of formwork components through processes such as cutting, assembling, and erecting. Plywood and lumber are particularly relevant to wood-based formwork because they can be prepared and assembled according to the required dimensions of concrete elements. The dimensional adjustment of these materials during formwork preparation can result in pieces that remain after the required sections have been cut. The generation of formwork offcuts is particularly evident in the preparation of plywood for slabs and beams. Bandara et al. (2026) examined plywood usage and wastage in slab and beam

construction and identified cutting as a major source of plywood offcuts. Their findings indicated that plywood wastage was influenced by factors such as structural geometry, material dimensions, handling practices, and construction-site conditions. These findings demonstrate that the preparation of plywood according to the dimensions of structural elements can produce offcuts of different sizes and quantities. Timber formwork waste has also been examined in relation to the materials and processes involved in concrete construction. (Hao et al., 2024) specifically investigated timber formwork waste and examined its generation within construction activities. Their study demonstrates that timber formwork can become a source of construction waste during the preparation and use of formwork systems. This is relevant to the present study because the use of timber-based formwork materials in residential construction can result in leftover pieces generated from cutting and dimensional adjustments. The generation of formwork offcuts may therefore occur during several stages of formwork preparation and installation, particularly during measuring, cutting, trimming, fitting, and dimensional adjustment. The quantity and characteristics of the resulting offcuts can vary depending on the original material dimensions, structural geometry, and requirements of the formwork components.

#### *2.3.2.1 Materials Commonly Used in Formwork*

Formwork systems are constructed using different materials depending on the requirements of the concrete structure, construction process, structural geometry, and desired surface finish. Li et al. (2022) explained that traditional formwork systems are commonly fabricated using rigid materials, particularly wood and metal. These materials are prepared through processes such as cutting, assembling, and erecting to create the required form for concrete elements. The choice of material is therefore associated with the physical

requirements of the formwork and the characteristics of the concrete structure being constructed. Wood-based materials are among the traditional materials used for formwork. Timber can be used as a formwork material because it can be cut and assembled according to the required dimensions and geometry of concrete elements. (Ahmad Shah, Noor, Kueh & Tamin, 2019) discussed wooden formwork for concrete casting and identified its application as a traditional approach to forming concrete members. (Li et al, 2022) likewise identified wooden formwork as one of the commonly used traditional formwork systems and noted that its fabrication involves manual processes such as cutting, assembling, and erecting formwork members. Plywood is another important wood-based material used in formwork. Plywood can function as the sheathing or contact surface against which fresh concrete is placed. Its sheet form allows it to be cut into dimensions appropriate for slabs, beams, columns, walls, and other structural components. (Bandara et al., 2026), in their study of plywood usage in slab and beam construction, examined the use of plywood in formwork and documented plywood wastage associated with the preparation and cutting of the material. Their findings demonstrate that plywood dimensions are directly relevant to formwork preparation because sheets may need to be adjusted to correspond with the dimensions and geometry of concrete elements. Lumber or timber members may also be used together with plywood as part of wood-based formwork assemblies. Lumber can serve as framing, supporting, or bracing components, while plywood can provide the surface that contains and shapes fresh concrete. The combination of these materials allows formwork components to be assembled according to the dimensions and configuration of individual structural elements. Hao et al. (2024) specifically examined timber formwork waste, demonstrating the relevance of timber-based formwork materials in discussions of construction material waste. Worku (2025) discussed plywood,

timber, steel, aluminum, and plastic as formwork materials and examined characteristics such as structural dimensions, load-carrying capacity, weight, safety, surface finish, cost, and construction requirements when considering formwork systems. The materials used in formwork can therefore be grouped into several major categories, including wood-based materials such as plywood and lumber. Each material has different physical characteristics and methods of fabrication and installation. For the present study, the focus will be placed on wood-based formwork materials, particularly plywood and lumber, because these materials are subject to cutting and dimensional adjustments during formwork preparation.

#### *2.3.2.2 Factors Affecting Formwork Material Utilization*

The utilization of formwork materials can be influenced by the requirements of structural elements, material properties, construction conditions, and methods of installation. Worku (2025) identified several factors relevant to formwork selection and use, including quality and surface finish, safety, structural dimensions, load-carrying capacity, weight, project duration, cost, and efficiency. These factors affect how formwork materials are selected and applied according to the requirements of a construction project. Material dimensions and structural geometry can also influence the amount of formwork material required during preparation. (Bandara et al, 2026), in their study of plywood used in slab and beam construction, found that plywood wastage was influenced by structural geometry, material dimensions, handling practices, and construction-site conditions. Similarly, Balasbaneh et al. (2024) discussed factors such as material characteristics, cost, construction requirements, and environmental considerations in the selection and analysis of formwork materials. These factors indicate that the quantity and characteristics of formwork materials used during

construction can vary according to project requirements and site conditions. For the present study, formwork material utilization will be considered in relation to factors such as structural dimensions, material dimensions, cutting requirements, material condition, handling practices, and construction-site conditions. These factors will provide a basis for documenting the conditions associated with the generation of formwork offcuts in two-storey residential buildings.

### 2.3.3 Causes of offcuts during construction

Offcuts are generated during construction due to various activities and conditions associated with material preparation and installation. Yu et al (2023) explained that off-cut wood is produced when timber is cut into required sizes, resulting in pieces with different dimensions and geometries. The resulting quantity and characteristics of these offcuts can therefore depend on the dimensions of the original material and the requirements of the construction activity. Improper cutting and dimensional adjustments are among the direct causes of material offcuts. Sweis et al. (2021) identified improper cutting of materials, design changes during construction, rework caused by labor mistakes, and poor site layout as major causes of material wastage. Design requirements can also contribute to offcut generation when material dimensions do not correspond with the dimensions of construction elements. A study on the root causes of construction waste found that frequent design changes, errors in design and construction details, and cutting materials into uneconomical shapes were highly important causes of waste generation. Bandara et al. (2026) likewise found that plywood offcuts in slab and beam construction were associated with cutting operations, structural geometry, material dimensions, handling practices, and construction-site conditions. For the present study, formwork offcuts are considered materials generated primarily from cutting, trimming, dimensional

adjustments, design requirements, and other construction-site activities. These causes are considered in relation to the generation of wood-based formwork offcuts during the construction of two-storey residential buildings.

### 2.3.4 Characteristics of construction offcuts

Construction offcuts may differ in their physical characteristics depending on the material, cutting process, and construction activity from which they are generated. Yu et al. (2023) documented off-cut wood using characteristics such as wood type, length, width, weight, and material proportions, while also noting that offcuts may have regular or irregular geometries. These characteristics provide measurable information for identifying and organizing individual offcut materials. The dimensions and geometry of an offcut are important characteristics when determining its potential for further use. Yu et al. (2023) noted that irregularly shaped off-cut wood can present greater difficulty because of its complex geometry. Similarly, Nasiri, Roy & Hughes (2025) characterized recovered wood based on form, dimensions, cracks, fixings, contaminants, useful cross-section, and useful length, demonstrating that both physical dimensions and material condition are relevant when determining the quality of recovered wood. The physical condition and presence of contaminants are also important characteristics. Faraca, Boldrin & Astrup (2019) found that wood waste varies according to its source, type, quality grade, and level of physical and chemical impurities. Their findings indicate that wood waste should be differentiated according to its individual material characteristics rather than treated as a single material category. For the present study, recoverable formwork offcuts will be characterized according to material type, length, width, thickness, shape, quantity, physical condition, and number of uses. These characteristics will provide the basis for documenting and classifying formwork offcuts generated

during the construction of two-storey residential buildings and determining which material characteristics should be recorded in the proposed inventory system. This shows that the dimensions of required formwork pieces, and how panels are cut to meet them, directly affect both waste and purchasing cost.

## 2.4 Formwork Cutting Optimization

Formwork cutting optimization involves planning the cutting of lumber and phenolic boards according to the dimensions required for concrete construction. The objective is to maximize material utilization and minimize cutting waste by selecting suitable cutting patterns. Mei et al. (2022) applied a two-dimensional cutting-stock model in a BIM-based formwork planning framework, while Wang and Yi (2022) developed optimization models to reduce offcuts from construction materials. These studies demonstrate how systematic cutting plans can reduce material waste.

### 2.4.1 Cutting patterns and dimensional requirements

A cutting pattern is the arrangement of required pieces taken from one stock piece, and the portion not assigned to any required piece becomes the offcut. Wang & Yi (2022) observed that many construction sites do not prepare a material processing plan, which leads to both surplus and waste. Their column generation method reduced waste by over 10% in some cases compared with two heuristic methods, and they noted the model could also apply to other one-dimensional materials such as wood. In the case of steel, Lee et al. (2020) generated cutting patterns from a rebar cutting list using minimum and maximum order lengths and a target loss rate, which lowered cutting waste. These studies show that the layout of required pieces on stock largely determines how much material is left over. For formwork specifically, Mei et al. (2022) noted that onsite formwork planning often depends on site managers individual experience, which

results in low efficiency and waste. Their BIM-based framework minimized the quantity of concrete forms using a two-dimensional cutting stock model based on a genetic algorithm. The required dimensions of formwork pieces therefore determine how panels are cut and how much is left over. The dimensions of what remains determine whether it can be used again. Rácz (2025) treated a leftover shorter than a set threshold as waste, so only longer leftovers retain value. Ma et al. (2025) studied a two-dimensional cutting stock problem with flexible length and usable leftovers, balancing the wasted area of material, the area produced beyond ordered quantities, and the number of machine adjustments. Senergues, Brahimi, Cherri, Klein & Péton (2026) reviewed 52 articles from 24 journals on the cutting stock problem with usable leftovers, categorized by dimension and planning period

### 2.4.2 Cutting waste minimization

Cutting waste minimization involves planning how construction materials are cut to meet required dimensions while reducing the amount of leftover material. In formwork construction, lumber and phenolic boards are cut to suit the dimensions of concrete elements. Poor cutting plans may generate offcuts that are too small or unsuitable for other applications. Therefore, planning cutting dimensions and identifying potential uses for leftover pieces can help improve material utilization and reduce unnecessary material purchases. Mei et al. (2022) developed a Building Information Modeling (BIM)-based framework for formwork planning that incorporated a two-dimensional cutting-stock optimization model using a genetic algorithm. The framework considered formwork requirements and potential panel reuse to minimize the quantity of formwork needed. Their illustrative case showed a 28% reduction in formwork purchase costs compared with expert-based planning. This study demonstrates how digital planning and cutting optimization can support more efficient formwork use and reduce material

costs. The most common approach to reducing cutting waste is the cutting stock problem. Wang and Yi (2022) developed an optimization model for one-dimensional construction materials that determines cutting patterns to minimize offcuts. Their study indicates that efficient cutting patterns reduce material losses by considering the available material dimensions and the required dimensions, and that material efficiency is tied to matching construction requirements with available stock. In reinforcing steel, Kwon et al. (2021) reviewed 52 studies on rebar cutting waste minimization and concluded that although many optimization algorithms have been proposed, a small amount of cutting waste is still generated in practice. Optimization does not remove leftovers completely, and the pieces that remain are generally short. Rácz (2025) defined trim loss as the leftover that is not long enough to be reused in the future, and in the proposed model a leftover shorter than a set threshold is counted as waste. This shows that whether an offcut is useful depends on its remaining dimensions as well as on how it was cut. Lee et al. (2020) also observed that leftover small-diameter bars may serve secondary purposes, but this is a minor use rather than direct reuse for the original application.

### 2.4.3 Material utilization in formworks

Material utilization in formwork cutting refers to how much of the plywood or lumber purchased ends up as the pieces required by the formwork, and how much is left over as offcuts. Bandara et al. (2026), in their study of plywood usage and wastage in slab and beam construction, identified cutting as a major source of plywood offcuts. They found that wastage depends on structural geometry, material dimensions, handling practices, and site conditions. Kern, Amor, Angulo & Montelongo (2018) used a multiple regression model on 22 high-rise residential buildings and found that wood waste was explained by the number of floors, the volume of in situ concrete, the site wood fence, and

the wood use rate. They also reported that limiting wood to formwork and avoiding other uses could reduce about 75% of wood waste. The scale of formwork waste shows why utilization matters. (Cheng et al., 2022) noted that formwork waste accounts for as much as 20–30% of total construction waste, and in their case study building, prefabrication reduced formwork waste by 59.68–71.83%. (Hao et al., 2024) described timber formwork as a crucial source of construction waste and quantified how timber formwork waste changes under different prefabrication rates. Together, these studies indicate that utilization in formwork cutting depends on the geometry and dimensions of the elements, how cuts are planned, and how well existing pieces are reused.

## 2.5 Construction Material Reuse and Recovery

The concept of construction material reuse and recovery have been progressively important to the development of the circular economy in the construction industry unlike the traditional concept of consume and dispose, material reuse seeks to maximize the value of material as long as possible implementing the goal of sustainable construction. Benachio et al., (2020) emphasizes that circular economy concept in construction projects involves strategies intended to eliminate the concept of disposal after use of an item and further promote strategies including the reuse, recycling and resource efficiency of materials. The main purpose of the circular economy is to reduce construction waste generation and resource extraction. Ginga et al., (2020) similarly emphasizes material recovery as an important track for management and reduction of construction and demolition wastes. Recovery of materials allows materials to be reused for other construction activities rather than just disposing of them. The study of Bellini, Andersen, Klungseth & Tadayon, (2024) focuses on implementing a circular economy to reduce the waste implication brought by construction activities. They studied the reuse of construction materials and

demonstrated that reuse does not only depend on the recovery of materials but also in efficient data management of those materials. These studies further reinforce that construction materials can potentially be used after their first use. Formwork materials can therefore be considered for potential recoverable resources when they retain suitable dimensions, number of uses and physical conditions.

### 2.5.1 Reuse of recoverable materials

Recoverable materials are material that can be potentially used after their first or original application such as formworks material. The study of Iacovidou, Purnell & Lim, (2018) integrated the use of smart technology to construction material reuse utilizing RFID as a means of tracking the information about the construction components. Their study highlights that lack of information about the construction components can contribute to ineffective reuse of components and that integrating tracking technology can support the reuse of construction materials. Reuse of recoverable materials often depend on the information regarding physical properties of the material (Byers et al., 2024). Ranasinghe et al., (2024) further establish that material data traceability information is important throughout the lifecycle of a material. In the study of Pronk et al., (2022) they demonstrated the reuse of timber formworks in a construction project involving discarded timber formworks. The study highlighted that timber formwork can be reused in other construction applications. These studies emphasize that recovered material should not simply be discarded or stored without documentation, Materials that are not properly documented can be misplaced and difficult to locate, and eventually be discarded and become waste.

### 2.5.2 Factors affecting construction material reuse

Construction material reuse is affected by factors such as material quality, physical condition, dimensions and number of uses. Tennakoon, Chileshe & Rameezdeen, (2022) their study identified that price, quality, market availability, information, standards affects the uptake for reprocessed construction materials. Among these material quality and physical condition is important for formwork reuse since formwork can be exposed to repeated loading and nail holes and other form of deterioration it is important to consider its condition before the reusing. Aramaki, Nakata, Otsuka & Miyata, (2021) investigates the changes in the quality of formworks and how it is reused, the properties including bending strength, stiffness, roughness and concrete surface characteristics were examined in their study. Their study demonstrated that repeated use of formwork affects its physical condition and therefore there is a need for evaluating before considering the reuse. Pronk et al., (2022) also found that the reuse of timber formwork can save material procurement and reduce waste. However, the reuse of materials may add complexity to the design and planning of construction activities. Balasbaneh et al., (2024) studied different formwork materials and their life cycles. They identified factors that can be considered in reusing of formworks. These studies emphasizes that factors affecting the reuse of construction materials are physical conditions, technical conditions, economic conditions and information or management. Physical conditions include the material condition, damage, deterioration and number of uses. Technical condition refers to dimensions, required performance, and compatibility with the construction requirement. Economic conditions include material cost, labor cost, storage and preparation of materials. Lastly, Information and management conditions are identification, traceability, inventory, availability of material records and storage of construction materials

that have potential for reuse in construction activities (Ranasinghe et al., 2024). These factors indicate that effective reuse of materials should consider these factors and not merely identify that a material exists in the inventory. The system should provide sufficient information and data about the material for it to be considered for potential reuse for future construction activities.

### 2.5.3 Construction material recovery and waste minimization

Material recovery is an important part of construction waste reduction because it aims to create a circular loop to maximize the value of a material before it is disposed. The study of Ginga et al., (2020) highlights that material recovery is one key component for a circular economy which potentially reduces the amount of construction waste dumped in landfills. Similarly, Osuizugbo, Awuzie & Oyeyipo, (2026) further highlighted that resource recovery is an important strategy for circular economy and reuse of materials, they also indicated that resource recovery is progressively being connected with digital technology and information based management of construction materials. Kabirifar et al., (2021) emphasizes that as the population grows and requires more construction because of urbanization the importance of the (3R) Reuse, reduce and recycle must be implemented for the management of wastes generated. The main denominator of waste management generally places prevention and reuse before recycling and disposal of materials. This is important for formworks because formworks can potentially be used several times before disposal maximizing its material value (Balasbaneh et al., 2024). Circular economy and waste material management have direct relationship with each other. Material reuse can contribute to the reduction of waste and conservation of resources leading to sustainable construction (Schützenhofer, Kovacic, Rechberger & Mack, 2022). The study of Hao et al., (2024) demonstrated a way to reduce the environmental impacts of formwork through prefabrication; their

study found out that prefabrication reduces environmental impacts. Similarly, Cheng et al., (2022) utilized building information modeling (BIM) for prefabrication and potential reduction to formwork wastes. Their study supports that digital information can support data management, quantification and evaluation of formwork waste reduction. These studies emphasized that formworks contribute to waste generated and consumes massive amounts of raw materials since reinforced concrete structures use formworks during construction. Construction industry must implement ways to reduce the construction waste generated during construction activities and reduce the consumption of raw materials by implementing circular economy and leading to sustainable construction (Hossain et al., 2020)

### 2.5.4 Digital information, technology and material traceability for reuse

The development of technology progressively influences the construction industry supporting design and construction activities Sacks, Girolami & Brilakis, (2020).BIM, RFID, database, mobile applications and other digital tools can provide aid for the construction industry. Byers et al., (2024) studied technologies such as data sheets, computer aided design (CAD), BIM, artificial intelligence and IOT systems for data management and bridging of information gaps for building materials. Their study emphasizes that successful reuse of material requires reliable information about the material properties, sources, condition and possible reuse information. Iacovidou et al., (2018) examined RFID and BIM for construction component reuse which demonstrated that tracking systems can retain the life cycle information and support reuse decision of construction components. Ranasinghe et al., (2024) further emphasizes that data of materials is crucial for traceability of building material circularity, they also identified information, design and technology as an important part for improving material

circular economy. These studies emphasize that the integration of technology and construction increases the progression of sustainable construction, it also gives importance to digital information and material traceability for the reuse of materials.

## 2.6 Formwork Reuse and Utilization Planning

Formwork utilization requires systematic planning to ensure that available materials are properly assigned to succeeding construction activities. Proper utilization planning can help minimize unnecessary delays and resource use while supporting the timely completion of construction work. In construction sites, different activities require specific spaces, equipment, materials, and access routes. When these resources are not properly coordinated, conflicts and inefficient use of available resources may occur, which can affect productivity and project scheduling (Whitman, Deshpande, Zech & Perez, 2021).

### 2.6.1 Formwork reuse in concrete construction

Formwork is a temporary construction system used to contain and support freshly poured concrete until it develops sufficient strength to retain its required shape. In contrast to permanent construction materials, formwork is then removed after concrete placement and can be reused for subsequent concrete work. This characteristic makes formwork an important material to consider for reducing construction waste and managing resources. In the study of Huszár & Lublóy, (2021) they examined six types of buildings derived from 22 finished buildings and compared the cost of formworks to the total cost of the construction project.

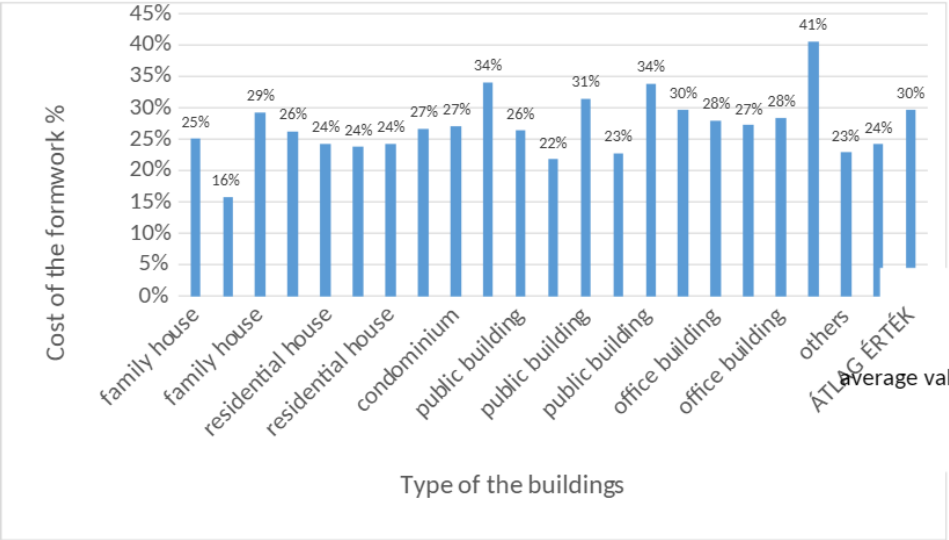



*Figure 2.1*:Proportion of the total formwork costs of all examined buildings related to the whole building structure

The conducted study found out that the residential buildings average formworks cost constitute to 24%-25% of the whole structural cost. In addition, Mei et al., (2022) found that formwork reuse across a construction supply chain resulted in a 25.38% cost saving, a 33.56% reduction in formwork quantity, and a 73.18% reduction in surplus reuse value compared with their study's baseline scenario. These findings demonstrate the potential economic benefits of systematically managing and reusing available formwork materials. These findings reinforced that formwork is significant and there is a greater importance in considering them for reuse to maximize its value. In recent years, greater attention has been given to the reuse of concrete formwork as a means of improving material utilization and reducing construction-related waste. Mei, Xu, Luo, Tan & Li, (2022) emphasized that concrete formwork has significant reuse potential, particularly when reuse is incorporated into construction planning and resource management. Rather than treating used formwork as a material that has reached the end of its use after only one application, reuse allows the same materials to

serve subsequent construction requirements when they remain suitable for use. The reuse of formwork can provide both economic and material-related benefits, particularly by reducing the demand for new formwork materials and maximizing the utilization of existing resources. Mei et al., (2022) investigated concrete formwork reuse within a construction supply chain and examined how formwork could be utilized again under changing construction requirements. Their study developed a mathematical model to analyze formwork utilization at individual construction sites and an agent-based model to simulate the dynamic reuse process. The results demonstrated that planned reuse can reduce the quantity of formwork required and improve the utilization of available materials. This suggests that formwork reuse can be incorporated into construction planning as a strategy for reducing unnecessary material consumption while making better use of resources already present on the site. The potential benefits of formwork reuse are also related to the number of times the material can be utilized. Balasbaneh et al., (2024) evaluated reusable formwork materials, including plywood, timber, steel, and plastic, based on their environmental and economic performance. Their analysis considered different numbers of reuse cycles and showed that the performance of reusable formwork materials can vary depending on how many times they are reused. This indicates that formwork reuse involves not only retaining previously used materials but also determining how they can continue to provide value through subsequent construction applications. Aramaki et al., (2021) further examined the quality changes associated with the reuse of plywood for concrete formwork at an actual building construction site. Their study investigated plywood reused up to four times and examined properties such as bending strength, bending stiffness, and surface roughness. The study demonstrates that repeated use can be associated with changes in material properties, supporting the need to consider the quality of previously used

formwork when determining whether it can be reused. Formwork reuse also requires planning because construction activities and material requirements may change throughout a project. Mei et al., (2022) highlighted the complexity of formwork reuse planning in a dynamic construction environment, where the availability and requirements of formwork may change over time. Similarly, Balasbaneh et al., (2024) demonstrated that repeated use can influence the overall environmental and economic performance of formwork materials. These findings indicate that effective formwork reuse requires organized information regarding available materials and their potential for subsequent use.

### 2.6.2 Factors affecting formwork reuse

The reuse of formwork is influenced by several factors that determine whether previously used materials remain suitable for subsequent construction activities. These include the type of formwork material, its condition, dimensions, and the number of times it has been used. Balasbaneh et al., (2024) examined plywood, timber, steel, and plastic formwork and found that their environmental and economic performance varied depending on the number of reuse cycles. This indicates that both the material type and its potential for repeated use are important considerations when determining whether formwork can continue to be utilized. The condition and quality of formwork are also important in determining its suitability for reuse. Pronk et al. (2022) identified uncertainty regarding the properties and quality of reclaimed materials as a challenge in material reuse. Their study on reused timber formwork demonstrated the need to examine the characteristics and condition of recovered materials before applying them to another construction use. This suggests that formwork should be evaluated based on its remaining quality rather than being considered reusable solely because it has been previously used. The dimensions and quantity of available formwork can further affect its potential for reuse. Pronk et al.,

(2022) noted that reclaimed materials may be available in different sizes and quantities, which can make it difficult to use them for a particular application. The available dimensions need to correspond with the requirements of the intended construction activity. Therefore, recording the dimensions and quantity of reusable formwork can help determine whether available materials can be appropriately utilized for succeeding formwork requirements. The number of previous uses is another factor that should be considered when evaluating reusable formwork. Balasbaneh et al., (2024) showed that the performance of formwork materials can change as the number of reuse cycles increases. This highlights the importance of considering previous use when determining the continued utilization of formwork. Overall, material type, condition, dimensions, quantity, and number of uses are relevant factors in identifying formwork that may remain suitable for reuse. These factors provide a basis for organizing information about reusable formwork and supporting its utilization in succeeding construction activities.

## 2.7 Construction Material Inventory Management

Inventory management refers to the systematic process of recording, organizing, monitoring, and controlling materials available for construction activities. It aims to maintain accurate information about available resources so that materials can be identified and used when needed. In construction projects, proper inventory management is important because materials must be available at the required time and location. Effective inventory control can help reduce stock shortages, unnecessary purchases, and excessive inventory costs (RathinaKumar, Lalitha Priya, Prasanna Kumar & Ravekumar, 2018). Construction projects involve a large number of materials that are delivered, stored, transferred, and consumed at different stages of the project. Because material requirements change as construction progresses, maintaining

updated inventory information is important for determining which resources are currently available. RathinaKumar et al., (2018) discussed the use of inventory control techniques to monitor construction materials and improve material availability. Their study indicates that organized inventory practices can assist construction personnel in determining material requirements and managing existing resources more effectively.

### 2.7.1 Material tracking and recording

Kasim, Kusumaningtias & Sarpin, (2019) explained that material tracking provides information about the availability of materials throughout construction activities. Their study noted that traditional material tracking practices can make it difficult to obtain timely and accurate information because materials may be distributed across different areas of the construction site. Without an organized tracking process, personnel may spend additional time searching for materials or may be unable to determine whether a required resource is already available on site. The recording of material information is particularly important because construction materials may be stored in different areas and may be required at different stages of the project. Kasim et al., (2019) reported that improper handling and storage can make materials difficult to track and locate when they are needed for construction work. Their findings also identified problems related to manual tracking, paper-based documentation, and the lack of timely information on material status. These issues can make it more difficult for construction personnel to determine whether materials are available and where they are located. For reusable formwork and construction offcuts, material tracking and recording can provide information needed to identify and utilize materials that remain available on site. Recording details such as material type, dimensions, quantity, and condition makes it easier to determine what materials are available. This information can also support the identification of possible uses

for recovered materials in succeeding construction activities. Therefore, a systematic tracking and recording process can help organize material information and support more efficient utilization of existing construction resources.

### 2.7.2 Inventory control on construction sites

Inventory control doesn't just have to be in construction but also in any businesses. Having to work in the corporate sector values the efficiency of inventory management (Nanaware & Saharkar, 2017). Moreover, Nanaware & Saharkar, (2017) discussed inventory control techniques that can be used to maintain appropriate material stock levels and support material resource planning. Their study examined techniques such as ABC classification and Economic Order Quantity (EOQ), which can help identify materials that require closer control and determine suitable inventory quantities. These techniques are intended to avoid excessive stock while reducing the possibility of material shortages that may interrupt construction activities. Material shortages and excessive inventory can both create difficulties on construction sites. RathinaKumar et al., (2018) applied ABC classification and EOQ analysis to construction materials and found that these techniques can help maintain sufficient stock, reduce inventory holding costs, protect stored materials from damage, and address stock-out problems. Their findings also showed that simple inventory control techniques can reduce inventory expenditure without requiring costly inventory management software. Lu, Wang, Xie & Wang , (2018) likewise examined material allocation decisions by considering inventory replenishment, material demand, project schedules, and cost. Their study demonstrated that decisions regarding when and where materials should be allocated are connected to both construction scheduling and inventory control. Nwankwo, (2023) identified material shortages, wastage, damage, storage constraints, and difficulties in locating materials among the challenges affecting construction

material inventory management. The study further examined approaches for improving material tracking and inventory management across construction projects. These concerns show the value of maintaining updated records of material quantity, location, and status. For recoverable formwork materials, inventory control can extend this practice by recording available pieces according to material type, dimensions, quantity, and condition. Such records can help personnel determine which existing materials can still be used for succeeding construction activities before additional materials are purchased. Inventory records can also support decisions concerning the movement and utilization of materials on site. Nor Azman, (2024) developed an inventory management system for construction materials, with the study focusing on inventory management and issues related to construction material delays. Similarly, recent research on construction material management systems has emphasized the value of providing updated information on material availability, movement, and stock levels to support timely decisions. For a construction site, maintaining this information in one inventory can make existing materials more visible to site personnel and provide a basis for deciding whether a recovered material can be utilized in another activity.

### 2.7.3 Problems with manual or informal material tracking

Manual and informal methods of material tracking remain common in construction projects, particularly in the recording, identification, counting, and monitoring of materials. Kasim et al., (2019) found that construction projects use different manual practices, including material identification, calculation, documentation, and tracking at the site and storage yard. Their study identified several problems associated with these practices, including difficulties in obtaining timely information about material availability, tracking materials at the

storage area, updating material status, and ensuring timely ordering. These limitations can make it difficult for construction personnel to determine the current location and availability of materials when needed. Manual tracking can also require considerable labor and time because material information must be recorded and updated through human intervention. Kasim et al., (2019) reported that manual material tracking in construction projects was associated with time-consuming procedures, labor-intensive tracking, extensive paper-based recording and reporting, and difficulty obtaining up-to-date material information. The study further noted that delivery orders and logbooks were commonly used for recording, making the tracking process more complicated and limiting the availability of current material status information. The difficulty of maintaining accurate material information becomes more evident when materials are transferred, stored, or moved between different locations. Kereri & Turner, (2018) Noted that inadequate material tracking on construction sites can result in significant time, cost, and effort being spent on counting, documenting, identifying, and locating required materials. The study further described situations in which materials were left scattered across the site or were not properly recorded and returned after use, making it difficult for workers to locate materials when needed. These problems can contribute to inefficient material management and time and cost overruns. Manual counting can present another source of error in construction-related material management. Ablidas, Monsura, Ablidas, & Dela Cruz, (2019) examined the manual counting of reinforcing bars and found that the process was highly susceptible to error, required substantial worker effort, and could not readily meet the requirements of real-time counting. The study also noted that incorrect quantities can result in additional checking, handling, and wasted time. Their findings demonstrate how dependence on manual counting can affect the accuracy and efficiency of material quantity

monitoring. The problems associated with manual tracking are particularly relevant when material information needs to be updated frequently. Kasim et al., (2019) identified difficulties in tracking and updating material availability, locating delivered materials, and monitoring materials at the construction site and storage yard. These limitations indicate that relying on conventional records may make it difficult for site personnel to obtain a clear and current overview of available materials.

## 2.8 Material Tracking System

A material tracking system provides an organized method for recording, storing, monitoring, and retrieving information about construction materials. Digital technologies can support material tracking by creating structured records containing material characteristics, quantities, and other relevant information. Ranasinghe et al. (2024) identified comprehensive material data and data traceability as important requirements for effective construction material management, while Keles, Rios & Hoque (2025) discussed digital platforms and material passports as tools for organizing and managing construction-material information. Digital inventories can also improve the visibility and traceability of materials throughout construction processes (Yang, Li, Yu & Zhong, 2024). Recent studies further demonstrate the development of digital systems for construction-material information management. De Wolf et al. (2024) proposed a digital workflow that integrates material cataloguing, tracking, tracing, and matching, while Byers, Stengele & De Wolf (2025) examined digital data carriers such as QR codes and NFC for tracking construction materials. These developments demonstrate how material information can be digitally recorded and connected to physical materials. For the present study, the material tracking system will focus on the recording, and retrieval of formwork offcut information

generated during the construction of two-storey residential buildings. The proposed system will organize information such as material type, dimensions, quantity, and number of uses to provide an accessible and structured record of available formwork offcuts.

### 2.8.1 Reusable material and offcut database

A material database provides a structured repository where information about construction materials can be recorded and retrieved. Yu et al. (2023) developed a database for off-cut wood containing information such as wood type, length, width, weight, moisture content, density, and material proportion. Their work demonstrates how physical characteristics of off-cut materials can be organized into a digital database to support material information management. Material passports provide another approach for organizing material information. Dai, Liu, Zhang & Chen (2019) described material passports as digital repositories that contain information about building materials and their characteristics, while Keles et al. (2025) identified material passports and digital platforms as technologies that can support the documentation and management of construction-material information. More recent research has likewise emphasized the importance of comprehensive material data and structured digital records for construction-material management (Ranasinghe et al., 2024). Digital inventories can also incorporate information from different technologies and data sources. De Wolf et al. (2024) demonstrated how digital tools can support the cataloguing and documentation of construction materials, while Yang et al. (2024) developed a digital twin-enabled platform for collecting and managing material information to improve material visibility and traceability. Byers et al. (2025) further examined QR codes, NFC, and other data carriers as means of connecting physical construction materials with digital information. For the present study, the

database will serve as the central repository for formwork offcut information. Each recorded offcut may contain data on material type, length, width, thickness, quantity, number of uses, and other relevant characteristics. The database will organize these records so that available formwork materials can be easily identified when a material requirement is entered into the proposed system.

### 2.8.2 Material tracking and matching

Material tracking involves maintaining information that allows construction materials to be identified, located, and monitored through digital records. Ranasinghe et al. (2024) emphasized that data traceability is an important requirement for construction-material management, particularly because material information can become fragmented throughout the building lifecycle. Yang et al. (2024) demonstrated the use of digital technologies to provide visibility and traceability of construction materials by linking physical material flows with digital information. Material matching involves connecting available materials with specific construction requirements based on their characteristics. De Wolf et al. (2024) developed a digital workflow that includes material cataloguing, tracking, tracing, and matchmaking. Their study showed that digital databases can organize information about available materials and facilitate matching between material supply and potential construction requirements. Similarly, Olumo & Haas (2024) proposed a digital framework for sourcing building materials by collecting and analyzing information about available new and reclaimed materials. Recent research has further emphasized the role of digital databases and identification technologies in material matching. Byers et al. (2025) examined QR codes and NFC as data carriers that can connect physical materials with their digital records, while Byers et al. (2024) identified databases, digital platforms, and other digital technologies as tools for linking available materials with project requirements. These studies demonstrate the

increasing role of digital information systems in organizing material supply and connecting it with construction demand. For the present study, material tracking and matching will involve recording available formwork offcuts and identifying suitable offcuts based on specified material requirements. The proposed system will use characteristics such as material type, dimensions, quantity, and number of uses to search the database and provide corresponding formwork offcut options when a specific requirement is entered.

# Chapter 3

# METHODOLOGY

## 3.1 Methodological Framework

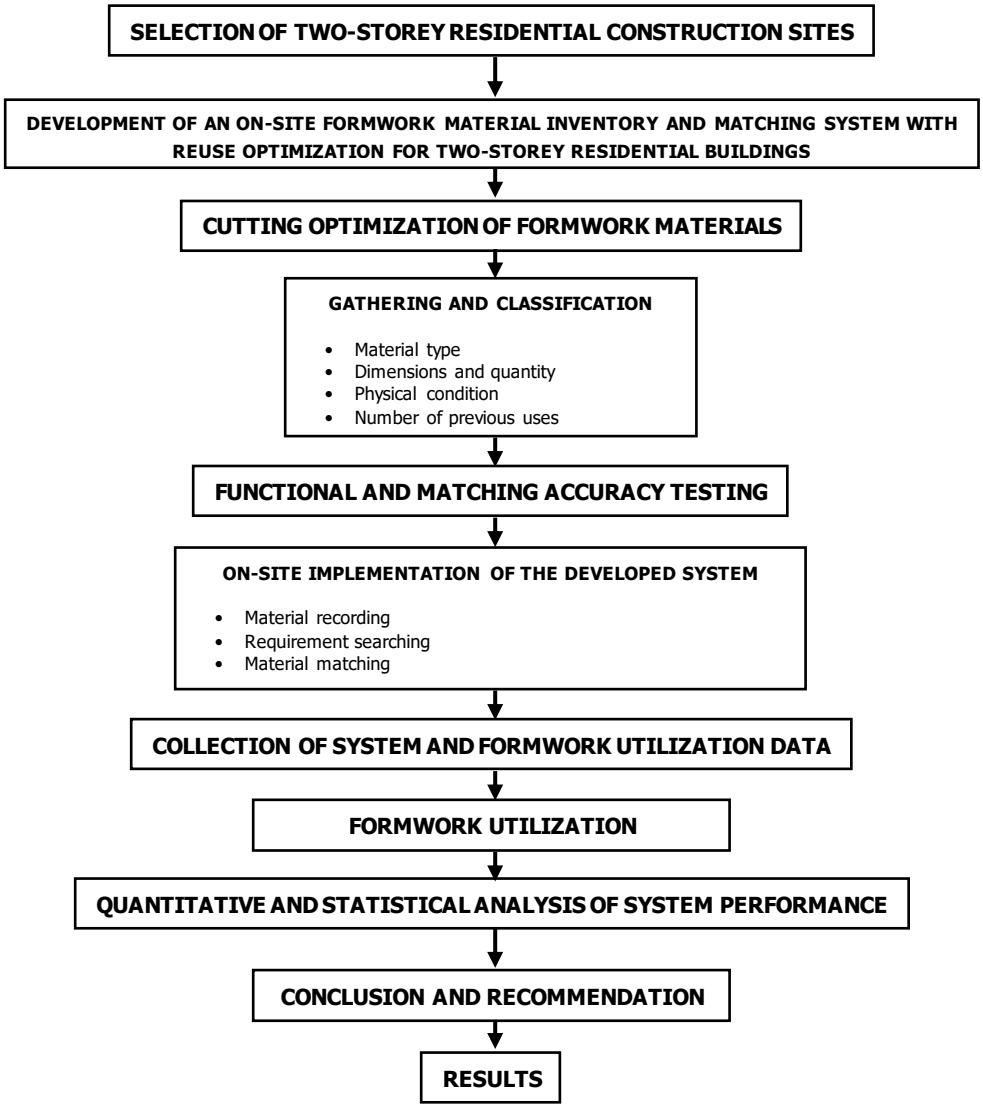



*Figure 3.1:* Methodological framework

## 3.2 Gathering of Data

The data that is required for conducting this study will be gathered from a selected 3 two-storey residential construction projects. The dimensions of beam, column and slab will be gathered for the formwork cutting optimization. The data collection will focus on formwork materials particularly lumber and phenolic board. The data gathering process will be conducted through construction site observation, physical measurement, material inventory, documentation of material dimension, number of reuse and condition. The purpose of the data gathering is to develop an accurate database for the available materials that can later be used by the system for inventory management and material matching.

### 3.2.1 Identification of formwork materials

The formwork materials available in the selected construction site will be identified. The study will include lumber and phenolic board materials used in formworks including new and used materials. Materials will be identified based on their type, dimension, quantity, number of reuse and condition. Formworks that are newly procured will be recorded as new and eligible for cutting optimization based on required dimensions and formwork material that are used will be recorded as reusable as it remains suitable for another formwork application.

### 3.2.2 Measurement of Material

Each recovered formwork material will be measured physically based on length, width and thickness before recording to the system. The dimension will be recorded in one uniform unit preferably in meters. Accurate dimensions are important in the system since this will be the basis for potential use in the future.

### 3.2.3 Identifying material condition

The physical condition of the formwork material will be assessed by inspection. Observing characteristics such as crack, splits, excessive deterioration, moisture damage, deformation and concrete residue will be examined.

## 3.3 Preparation of Materials

The required dimensions of beams, columns and slab will be determined for the cutting optimization. The identified recovered formwork materials will be gathered and examined by its existing condition. Materials that are suitable for another reuse will be separated from the materials that are damaged and excessively deteriorated. The material examination will consider visible defects such as cracks, split, deformation and deterioration. Each formwork material that is examined to be suitable for reuse will be measured physically. Concrete residue, dirt, nails will be cleared as long as it does not damage the material. The condition of the examined material shall be classified as new, and reusable based on their physical condition. Each recorded material will be assigned a unique identification number (ID).

## 3.4 Schematic Diagram

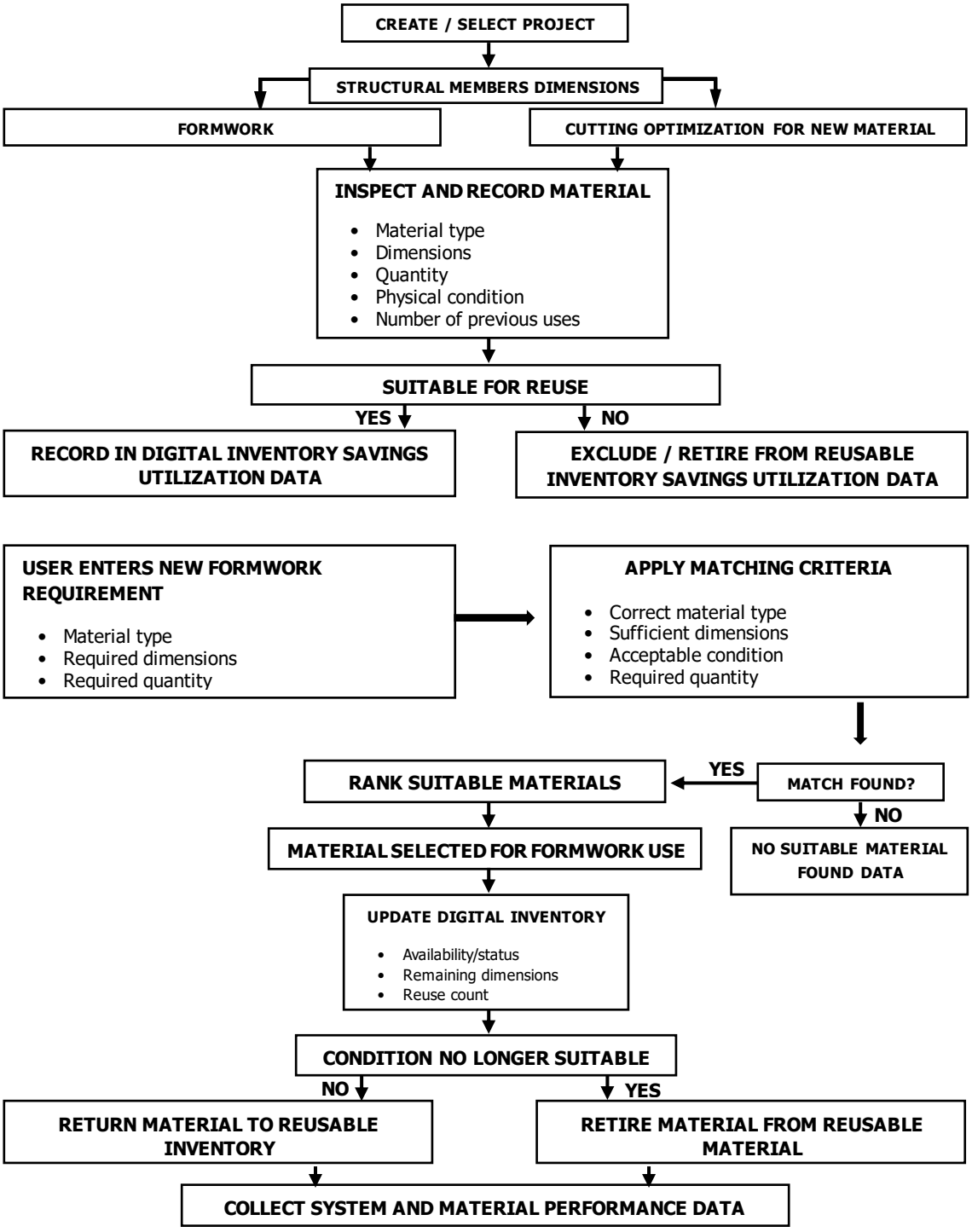



*Figure 3.2:* Schematic diagram

## 3.5 Procedure

The procedure of the study will be conducted sequentially, following the activities presented in the methodological framework. The procedure will cover the selection of two-storey construction sites, gathering and classification of formwork, development of the digital inventory system, testing and on-site implementation, collection of utilization data, and analysis of the system's performance.

### 3.5.1 Selection of two-storey residential construction sites

1. Identify potential two-storey residential construction sites that meet the scope of the study.
2. Screen the sites based on the availability of formwork materials, particularly phenolic board and related formwork materials.
3. Coordinate with the head project personnel or site management to obtain permission to conduct observations, collect material information, and implement the developed system.
4. Select the construction sites that satisfy the established criteria and can provide sufficient formwork materials data for the study.

### 3.5.2 Design and development of the digital formwork reuse utilization inventory and formwork optimization system

1. Determine the required system inputs and outputs based on the data gathered during the site assessment.
2. Design the inventory structure to accommodate records for material type, dimensions, quantity, condition, and number of previous uses

3. Develop the prototype as a Progressive Web App (PWA) that can be accessed through web-enabled devices.
4. Develop the material cutting optimization function
5. Develop the material inventory function for adding, updating, storing, and retrieving formwork and offcut records.
6. Develop the material search and filtering functions to allow users to locate available materials based on specified characteristics.
7. Develop the utilization function to identify potential uses for available formwork materials based on their recorded characteristics and required dimensions.
8. Incorporate material matching functionality to compare available materials with required formwork dimensions or material requirements.
9. Develop the reuse tracking function to record and update the utilization history of reusable formwork materials.
10. Test the developed system during the development stage to identify and correct errors before on-site implementation.

### 3.5.3 Cutting optimization of formwork materials

1. Determine the required system inputs of beam, column and slab dimensions
2. Develop the optimization of cutting for the new formwork material
3. Choose whether to use the formwork or store it in the inventory for later use

### 3.5.4 Gathering and classification of formwork data

1. Identify the formwork materials available at the selected construction sites.
2. Gather the required inventory information for each material, including material type, dimensions and quantity, physical condition, and number of previous uses.
3. Measure the dimensions of recoverable formwork materials using appropriate measuring tools.
4. Classify the materials according to their type, dimensions, condition, and potential for further use.
5. Record the number of previous uses of reusable formwork materials based on available site records or information provided by site personnel.
6. Establish a material record for each identified formwork material to facilitate tracking within the proposed system.

### 3.5.5 On-site implementation of the developed system

1. Deploy the developed system at the selected construction site.
2. Input the identified formwork materials into the system using the information collected during the site inventory.
3. Input the dimensions of beams, columns and slab
4. Choose whether to reuse a material or cut a new material
5. For new material the system will determine the optimized cutting for the required dimension

6. Update the material records when materials are reused, or otherwise changed in status.
7. For the reuse, use the requirement search and material matching functions to identify available materials that may satisfy the required dimensions or intended application.
8. Record the reuse of formwork materials and update their number of previous uses after each utilization.
9. Continue monitoring the materials throughout the implementation period to generate actual inventory and utilization data.

### 3.5.6 Quantitative and statistical analysis of system performance

1. Prepare material records and formwork requirements based on the identified material types, dimensions, quantities, number of previous uses, and physical conditions.
2. Enter the prepared material records into the developed prototype and verify whether the system correctly records and stores the information.
3. Test the material search and retrieval functions by entering selected material requirements and determining whether the corresponding inventory records can be located.
4. Test the matching function by providing predetermined formwork requirements and comparing the materials identified by the system with the manually verified appropriate matches.
5. Record the number of correct and incorrect recording and matching results to determine the prototype's recording and matching accuracy.

6. Test the utilization function by determining whether the prototype can identify potential uses for the recorded formwork offcuts and reusable materials.
7. Record the materials that are successfully identified for potential reuse or utilization and determine the corresponding utilization rate.
8. Analyze the results using frequency, percentage, mean, and other appropriate descriptive statistical measures to determine the performance of the prototype.
9. Summarize the findings to determine whether the developed prototype performs its intended functions as an on-site inventory, matching and cutting optimization system for formwork materials.

### 3.6 Instrumentation

The primary instrument of the study will be a Progressive Web Application (PWA) prototype developed as an on-site formwork material inventory, matching, and cutting optimization system for two-storey residential buildings. The application will serve as the main tool for recording, organizing, monitoring, retrieving, optimization and matching available formwork materials with project requirements. It will include an inventory recording module where users can enter relevant material information, including material type, dimensions, quantity, quality condition, and number of previous uses. These data will allow the system to maintain an organized record of available formwork materials. The prototype will allow users to input the dimensions and quantities of structural elements, such as beams, columns, and slabs. Based on the entered structural dimensions and predefined formwork calculation rules, the system will determine the required formwork materials, including their types, dimensions,

and quantities. The system will present these requirements as outputs and identify available inventory materials that match the specified requirements. For phenolic boards, the prototype will include a cutting optimization function that generates proposed cutting layouts based on standard board dimensions and the calculated formwork requirements. The output will include the proposed cutting dimensions, number of boards required, and estimated leftover material. To evaluate the prototype, the researchers will conduct functional testing using prepared material inventory data and test cases representing formwork requirements in two-storey residential construction. The inventory data will be entered into the prototype to determine whether the system can correctly record, store, retrieve, update, and organize material information. The researchers will test the structural dimension input function by entering the dimensions and quantities of beams, columns, and slabs and comparing the system-generated formwork requirements with manually verified calculations. The material matching function will be tested by comparing the materials identified by the system with manually verified matches based on the required material types and dimensions. The cutting optimization function will be tested by examining the feasibility of the generated cutting layouts, the number of boards required, and the estimated leftover material. The system-generated layouts will be compared with manually prepared layouts using the same requirements and standard board dimensions to determine the difference in cutting waste. A Prototype Evaluation Checklist will be used to assess the performance of the developed system. For the inventory management function, the criteria will include the ability to record, store, retrieve, update, and organize material information. For the structural dimension input and formwork calculation functions, the criteria will include the accuracy of the entered data and the calculated formwork material types, dimensions, and quantities. For the material matching function, the criteria will

include the ability to identify available inventory materials that meet the specified requirements. For the cutting optimization function, the criteria will include the feasibility of the generated cutting layouts, the number of boards required, and the amount of estimated leftover material. The results of the functional testing and checklist evaluation will be used to determine whether the developed PWA performs its intended inventory management, formwork calculation, material matching, and cutting optimization functions. Any identified errors, missing functions, or inconsistencies will be documented and considered in improving the prototype before the final version is presented.

### 3.7 Statistical Analysis

The data gathered in the study will be analyzed using descriptive statistics to organize, summarize, and interpret the results obtained during the testing and evaluation of the developed Progressive Web Application (PWA). The analysis will focus on the performance of the system in inventory management, formwork calculation, material matching, and cutting optimization. For inventory management, the frequency and percentage of accurate and inaccurate entries will be determined based on material type, dimensions, quantity, number of previous uses, and physical condition. The recorded information will be compared with the prepared material records to determine the accuracy of the system in storing and managing formwork inventory data. For formwork calculation, the dimensions and quantities of beams, columns, and slabs entered into the system will be used to generate the required formwork material specifications. The calculated material types, panel dimensions, and quantities will be compared with manually verified calculations. The frequency and percentage of correct and incorrect results will be determined to evaluate the accuracy of the system's calculations. For material matching, the number of correct matches identified by the system will be compared with the total number of tested matching cases.

The matching accuracy will be expressed as a percentage to determine how accurately the system identifies available inventory materials that meet the required material types and dimensions. For cutting optimization, the generated cutting layouts will be examined to determine whether the required phenolic panels can be accommodated within the specified standard board dimensions. The number of boards required, and the estimated leftover material will be recorded for each test case. The results will be presented using tables and graphical representations, supported by frequency counts and percentages. The mean will be used when appropriate for summarizing numerical results across multiple test cases. These descriptive statistical measures will provide a basis for evaluating the accuracy of inventory recording, formwork calculations, and material matching, as well as the performance of the cutting optimization function in generating feasible layouts and reducing estimated cutting waste.

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# APPENDICES

## Appendix A: Budget Plan

**BUDGET PLAN**

| ITEM | COST (PHP) | QUANTITY | UNIT | TOTAL (PHP) |
|---|---|---|---|---|
| **A. FIELD WORK** | | | | |
| Laborers/Site assistance | 3,000 | 1 | est. | 3,000 |
| Transportation | 4,000 | 1 | est. | 4,000 |
| **B. DOCUMENTATION** | | | | |
| Printing & Photocopying | 2,000 | 1 | est. | 2,000 |
| Binding | 220 | 1 | unit | 220 |
| **TOTAL** | | | | **9,220** |
| **TOTAL WITH 10% CONTINGENCY** | | | | **10,142** |

Note: All unit costs are estimated based on prevailing market rates as of 2026. The estimated cost of developing the inventory system was determined through consultation and discussion with a third-party application developer. The 10% contingency covers unexpected expenses and additional field visits.

## Appendix B: Gantt Chart

| CODE | Month | November | | | | December | | | | January | | | | February | | | | March | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Week | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 |
| | Activity | | | | | | | | | | | | | | | | | | | | |
| A | Development of System | | | | | | | | | | | | | | | | | | | | |
| B | Initial Testing of the Developed System | | | | | | | | | | | | | | | | | | | | |
| C | Troubleshooting of System Based on the Testing Results | | | | | | | | | | | | | | | | | | | | |
| D | Data Collection | | | | | | | | | | | | | | | | | | | | |
| E | Data Analysis and Evaluation | | | | | | | | | | | | | | | | | | | | |
| F | Writing of Chapter 4 | | | | | | | | | | | | | | | | | | | | |
| G | Writing of Chapter 5 | | | | | | | | | | | | | | | | | | | | |
| H | Submission of Chapter 5 for Adviser's Comment | | | | | | | | | | | | | | | | | | | | |
| I | Revision Based on Adviser's Comment | | | | | | | | | | | | | | | | | | | | |
| J | Finalization of Manuscript | | | | | | | | | | | | | | | | | | | | |
| K | Final Defense Preparation | | | | | | | | | | | | | | | | | | | | |
| L | Final Defense | | | | | | | | | | | | | | | | | | | | |